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1. Introduction

1.1 Purpose of Document

Purpose of document is to configure APM to Unisim Integration steps and any troubleshoot/common issues encountered for this interface

1.2 Basic Check steps

For any issue/error encountered in the this interface, follow below preliminary steps to troubleshoot.

1. Check the AM logs for any error specific to issue from the Calculation server at
<InstallDrive>:\ProgramData\Honeywell\AssetManager\LogFiles\AMLogs.
2. Follow the Troubleshooting guide provided with Standard APM documentation
3. Access the “Knowledge Article” through [://process.honeywell.com/us/en/services-and-support/support-center/technical-support](https://process.honeywell.com/us/en/services-and-support/support-center/technical-support)

Search the issue by filter “Product Family as Asset Management” and “Product as Uniformance Asset Sentinel”

4. If still issue is not able to find, check with respective project lead or else raise the SR.

2. UNISIM integration with APM

2.1 Pre-requisite:

2.1.1 Software Compatibility

UAS 540.1/.2/.3 version is compatible with Unisim R492, R500 and R510.

Note: Although R500 and R510 isn't tested by DE but project team has tested and it's working fine.

UAS Version	Unisim Version
R540.1	R492, R500, R510
R540.2	R492, R500, R510
R532.5	R492

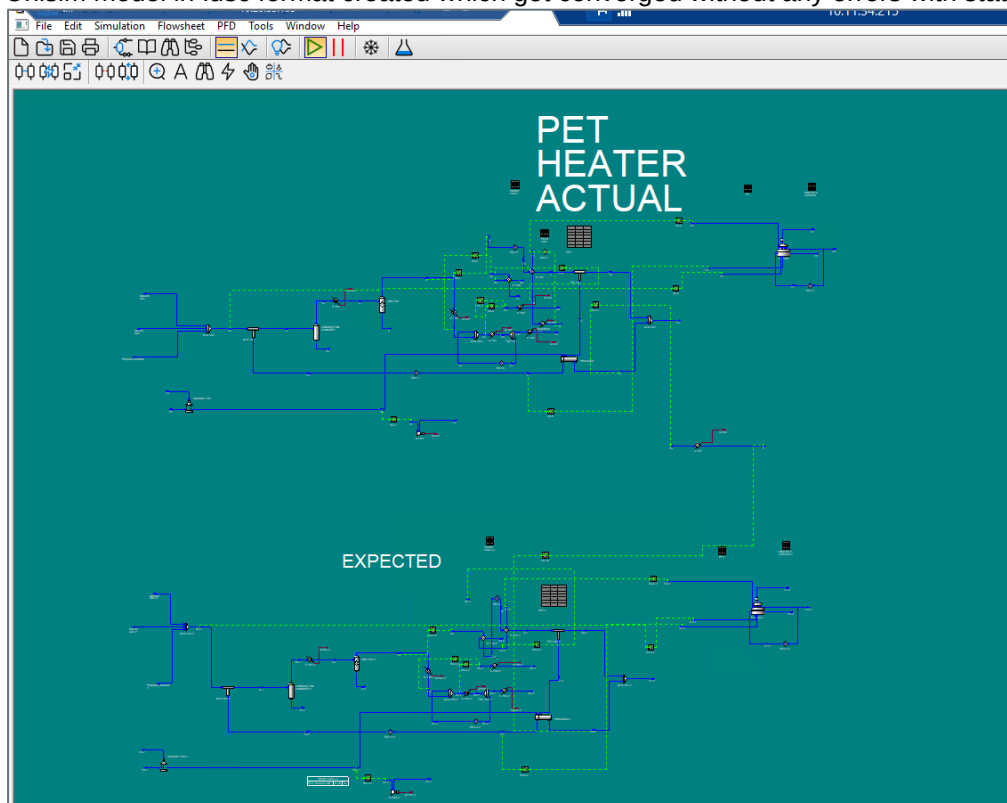
2.1.2 License

Ensure that APM unisim interface should have below license:

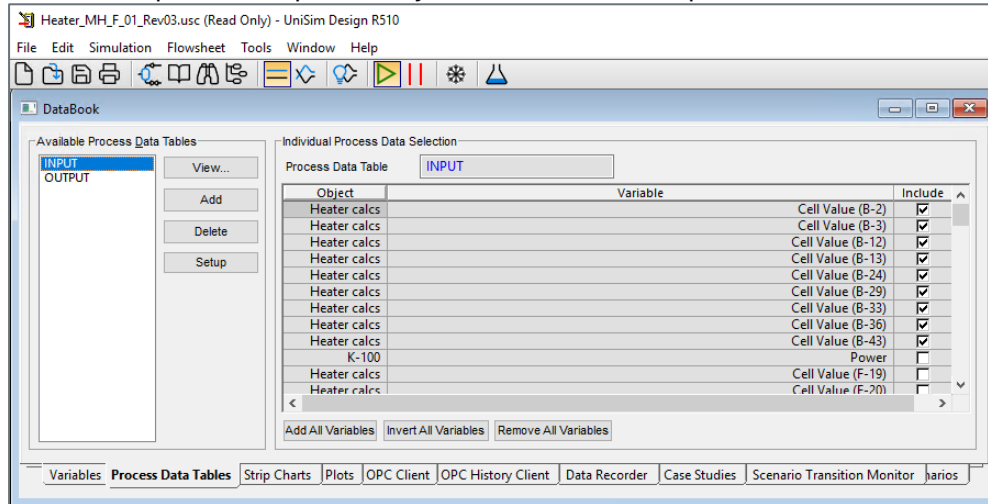
1. Sentinel.UniSim Link (SEN) – For APM to Unisim interface.
2. UniSimDesign.Rating (USD) – Required for Snapshot file creation for unisim case and maintenance of unisim case for any update.

2.1.3 Unisim Build Installation and case creation

1. Ensure that a licensed version of USD simulator (version R492/500/510) is installed on the Calculation server.
2. Unisim model in .usc format created which get converged without any errors with static data.



3. Ensure that tag map all the input variables configured in USD as "Inputs" and output variables configured in USD as "Outputs" to the Asset Performance Management attributes.
4. Ensure that the process data table names are configured for all input attributes written from Asset Performance Management as "Input" and output attributes read by Asset Performance Management as "Output" in the USD model.
5. Check the Input and Output PDT by "Ctrl+D" and select Input → View



6. Ensure that all Attributes should be mapped correctly with Access Mode as "Read/Write". There should not be any constant/fix value with No tag mapping and "No Transfer" as Access Mode.

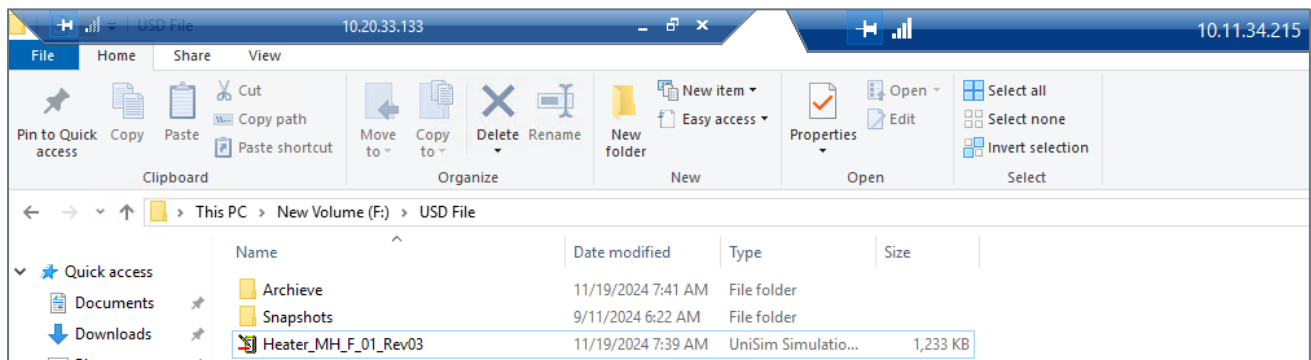
	Object	Variable	Value	Units	Tag	Access Mode
1	Heater calcs	Cell Value (B-2)	1.418e+004	kg/h	BASE_AIR_MASSFLOWRATE	Read/Write
2	Heater calcs	Cell Value (B-3)	1151		BASE_NG_FLOWRATE	Read/Write
3	Heater calcs	Cell Value (B-12)	798.4		BASE_FG_CONVECTIONINLET	Read/Write
4	Heater calcs	Cell Value (B-13)	431.3		BASE_FG_CONVECTIONOUTLE	Read/Write
5	Heater calcs	Cell Value (B-24)	338.8		BASE_PROCESS_OUTLETTEMP	Read/Write
6	Heater calcs	Cell Value (B-29)	312.7		BASE_PROCESS_INTLETTEMP	Read/Write
7	Heater calcs	Cell Value (B-33)	726.5		BASE_PRODUCT_FLOWRATE	Read/Write
8	Heater calcs	Cell Value (B-36)	7.138		BASE_PRODUCT_OUTLETPRES	Read/Write
9	Heater calcs	Cell Value (B-43)	189.8		BASE_AIR_EXITTEMP	Read/Write
10	Heater calcs	Cell Value (B-4)	0.0000		BASE_PROPAANE_FLOWRATE	Read/Write
11	Heater calcs	Cell Value (B-5)	178.2		BASE_WASTEGAS_FLOWRATE	Read/Write

NOTE All the variables that you select must be tag-mapped in Asset Performance Management. Hence, ensure that you select only the variables that you require.

OUTPUT Data						
	Object	Variable	Value	Units	Tag	Access Mode
1	K-100	Power	1.700	kW	OP_ACT_AIRFANPOWER	Read/Write
2	Heater calcs	Cell Value (F-19)	5.581	%	OP_ACT_EXCESSAIR	Read/Write
3	Heater calcs	Cell Value (F-20)	86.65		OP_ACT_OVERALLEFFICIENCY	Read/Write
4	FH-100	Total Duty	4.371e+007	kJ/h	OP_ACT_TOT_DUTY_PREDICT	Read/Write
5	Heater calcs	Cell Value (B-21)	10.67		ACT_AIRTOFUELRATIO	Read/Write
6	FH-100	Combustion Out Temp	2042	C	OP_ACT_FLAMETEMP	Read/Write
7	Heater calcs	Cell Value (B-32)	12.06	kJ/h	ACT_FUELDUTY	Read/Write
8	E-106	Exchanger Cold Duty	3.468e+007	kJ/h	OP_ACT_RADIATIONDUTY	Read/Write
9	E-102	Exchanger Cold Duty	9.038e+006	kJ/h	OP_ACT_CONVECTIONDUTY	Read/Write
10	Objective func	Cell Value (E-3)	945.8	C	OP_EXP_FLUEGASRADIATIONC	Read/Write
11	Objective func	Cell Value (E-5)	387.0	C	OP_EXP_CONVECTIONOUTTEI	Read/Write
12	Objective func	Cell Value (E-6)	339.5	C	OP_EXP_PROCESSOUTTEMP	Read/Write
13	Heater calcs-2	Cell Value (G-37)	1.386e+007	kJ/h	OP_EXP_CONVECTIONDUTY	Read/Write
14	Heater calcs-2	Cell Value (G-38)	3.158e+007	kJ/h	OP_EXP_RADIATIONDUTY	Read/Write
15	Heater calcs-2	Cell Value (G-39)	4.545e+007	kJ/h	OP_ACT_TOT_DUTY_PREDICT	Read/Write
16	Heater calcs-2	Cell Value (G-40)	104.0		OP_EXP_EFFICIENCY_PREDICT	Read/Write
17	Heater calcs	Cell Value (K-41)	1.006	mol %	ACT_STACKO2	Read/Write
18	Heater calcs	Cell Value (L-41)	1.145	mol %	EXP_EXPECTEDSTACKO2	Read/Write
19	Heater calcs	Cell Value (K-42)	3.227e+006	kJ/h	ACT_APHDUTY	Read/Write
20	Heater calcs	Cell Value (L-42)	3.124e+006	kJ/h	EXP_EXPECTEDAPHDUTY	Read/Write
21	Heater calcs	Cell Value (K-43)	292.9	C	ACT_STACKTEMP	Read/Write
22	Heater calcs	Cell Value (L-43)	251.6	C	EXP_EXPECTEDSTACKTEMP	Read/Write
23	Heater calcs	Cell Value (K-44)	0.3500		ACT_APHBYPASS	Read/Write
24	Heater calcs	Cell Value (L-44)	0.3500		EXP_EXPECTEDAPHBYPASS	Read/Write
25	Heater calcs	Cell Value (L-46)	1.741	kW	OP_EXP_AIRFANPOWER	Read/Write
26	Profit calc	Cell Value (C-21)	4.987e+004		USDPERYEARSAVING	Read/Write
27	Profit calc	Cell Value (C-18)	36.96	kg/h	FUELFLOWREDUCTION	Read/Write
28	Heater calcs-2	Cell Value (E-20)	90.09	%	OP_EXP_EFFICIENCY	Read/Write

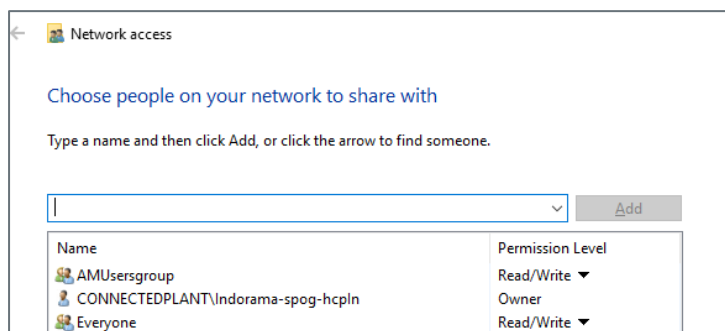
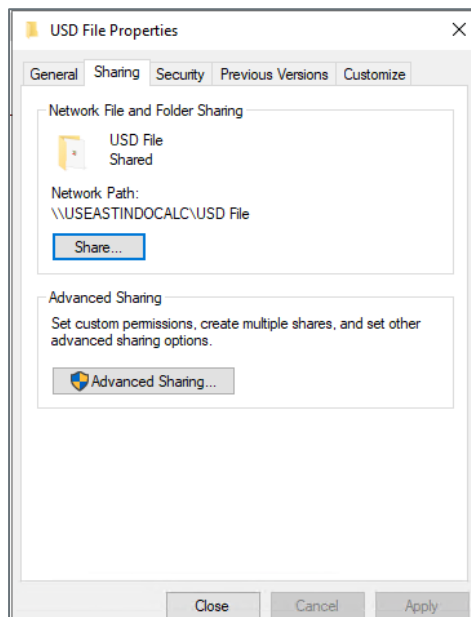
2.1.4 Security and Permission for Unisim Case

Unisim case should saved in the calculation sever in any folder of drive.

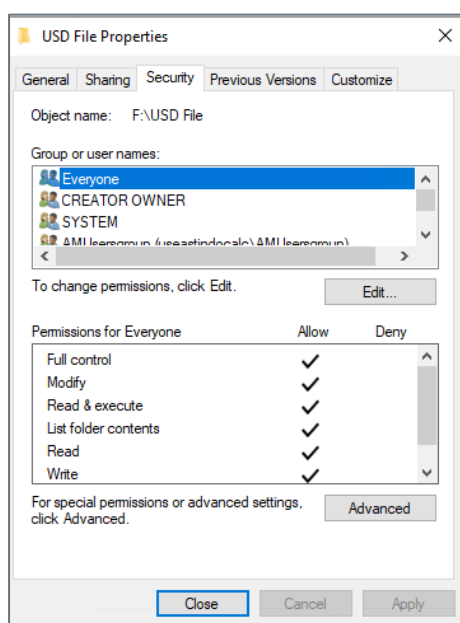


Follow below steps to ensure Unisim folder has right permission.

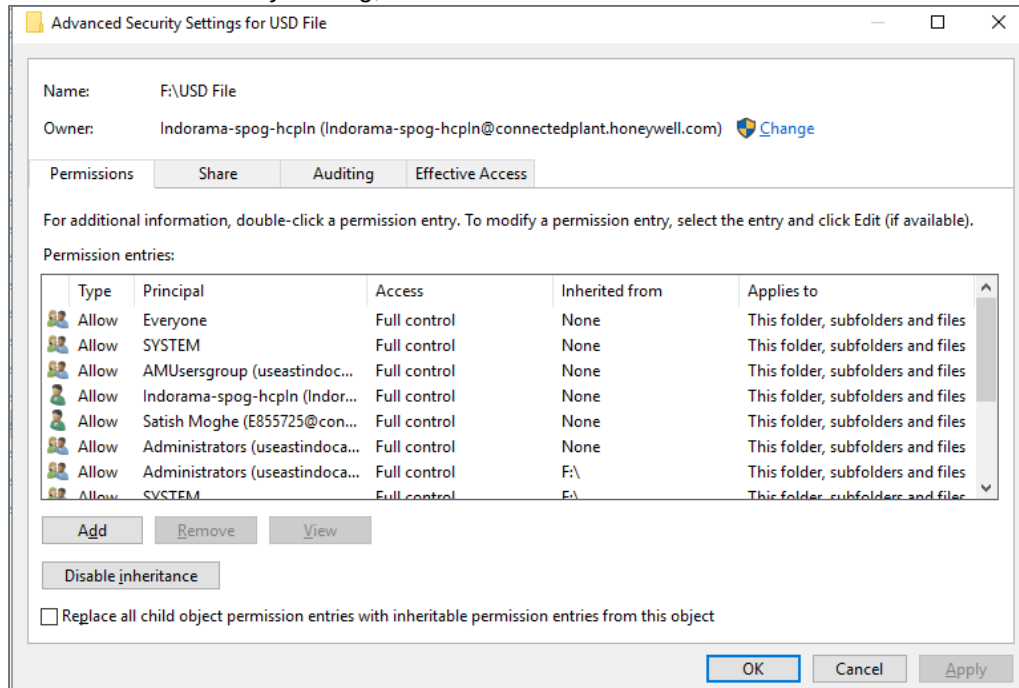
1. The Asset Performance Management services user must have permission to access the UniSim case file, set at the file level. If permission is not set, or is set at the folder level only, an error will result when the attempting to execute the UniSim model.
2. Unisim file folder should share for services user/Everyone.



3. Full control rights should be granted to “AMUsersgroup, Everyone, Service Account” from the Security tab of USD folder.

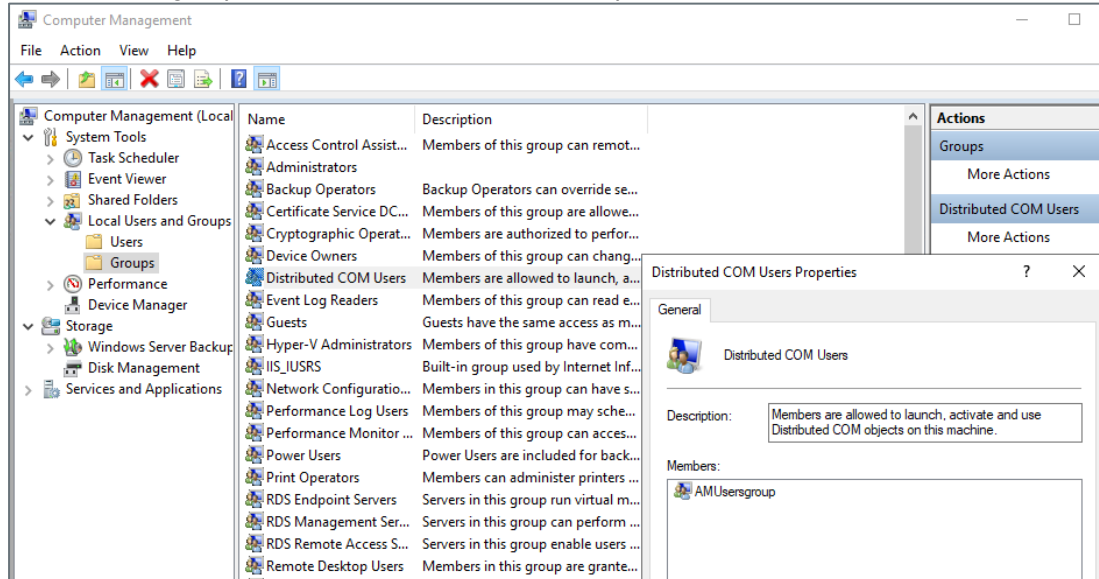


- From Advance Security Setting, USD file should **Enabled inheritance**.

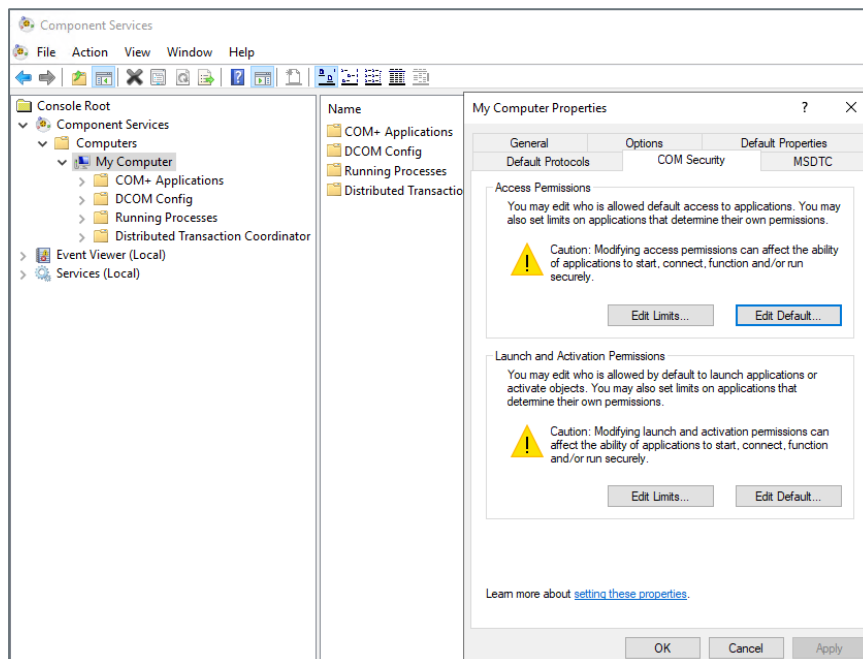


2.1.5 DCOM setting on calculation server:

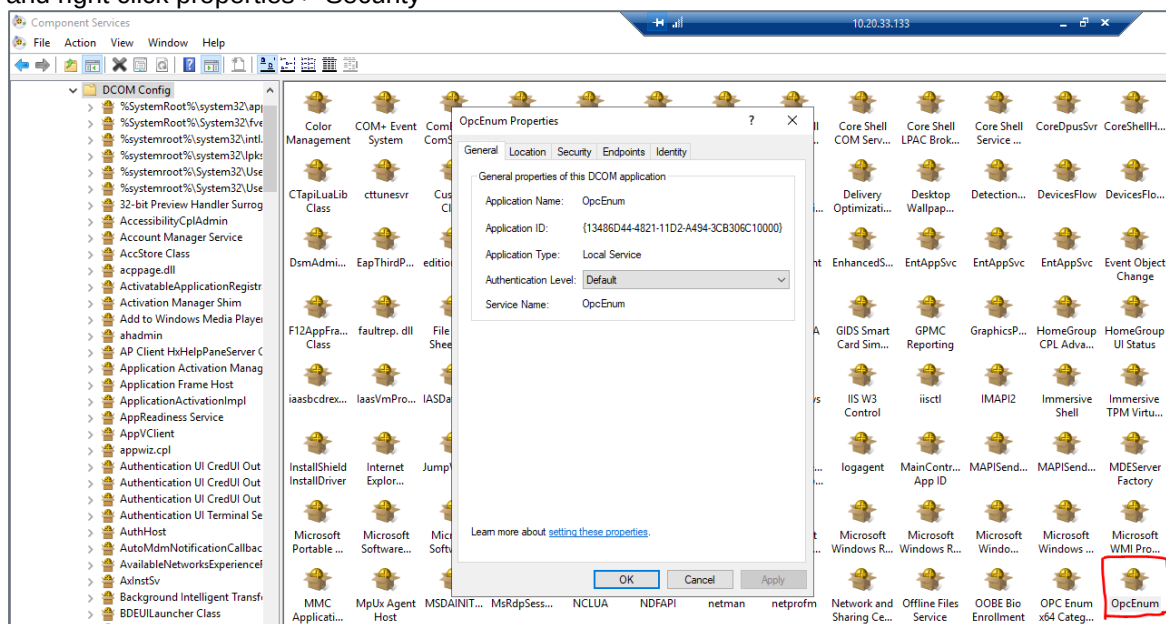
- Add AMUsersgroup in Distributed COM Users Group



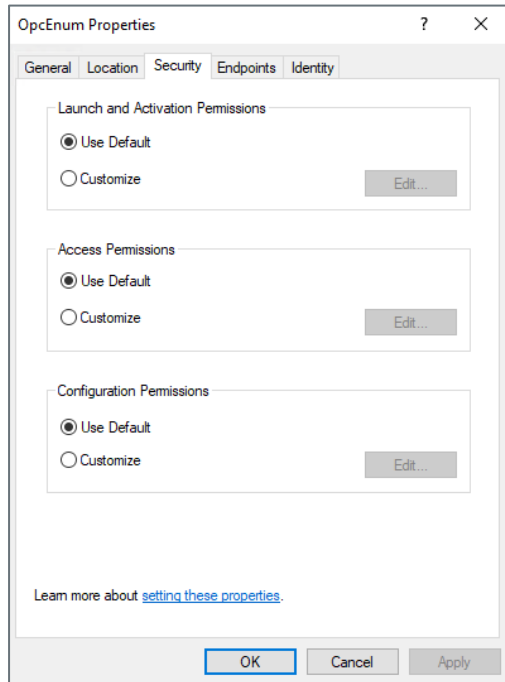
- Go to Start > All Apps > Component Services > Computers > My Computer and right click properties > COM Security



3. Add NETWORK in Access Permissions for Edit Limits and Edit Default
4. Launch and Activation Permissions for Edit Limits and Edit Default. Give all permissions for NETWORK.
5. Go to Start > All Apps > Component Services > Computers > My Computer>DCOM Config> OPCEnum and right click properties > Security



6. Mark “Customize” for Launch and Activation Permission, Access Permission and Configuration Permission.



2.1.6 To configure the UniSim Design data source

1. Open Asset Performance Management, and on the Contents tree choose Configuration > Datasources.
2. In the DataSource page, select click Add and select USD: UniSim Design. The USD data source Configuration pane is displayed in the right.
3. In the Data Source Name box, the name of the default data source is displayed. If you want to change the name of the default data source, type a new name for the data source.
4. In the Display Name box, type a name for the data source in the localized language. On the tree view, the name of the data source is displayed in the localized language.
5. In the Input File box, type the path where the USD case file is located.

NOTE: If you have a separate calculation server then the input folder and snapshot location is created in the calculation server. Create a shared network folder for the input file and snapshot location and update the folder path. Example: Input File [CalculationServer]\UnisimModels\USDModelUSC]

\\10.11.34.215\USD File\Heater_MH_F_01_Rev03.usc

Snapshot Location [CalculationServer]\UnisimModels\Snapshots\]

\\10.11.34.215\USD File\Snapshots

6. In the Snapshot Location box, type the file path to the folder in which to save the UniSim snapshot, which is the output of the UniSim model execution. Inside this folder location, the snapshots are saved to a Success or Fail folder, depending on the result.
7. In the Timeout box, type the maximum time in minutes the simulation model runs, after which the model is stopped.
8. From the Model list, select one of the following: | Calculation: the calculation of equipment/process performance using the configured UniSim design model. | Optimization: identifying the optimal process condition or equipment parameters using the configured UniSim design model.
9. Click Save to save the USD configuration.
10. Once datasource successfully get saved, Datasource will show as “Not Running”. It needs AMApplicationServer restart once Input and output tags in Asset Attribute mapping and Unisim has been done else it will not show “Running”.

2.1.7 Configuring Input/Output Variables in Asset Attribute Mapping sheet.

This section describes how to configure Unisim input/output variables with APM input/output variables in Asset Attribute Mapping Sheet.

The following are the high-level tasks to integrate USD with Asset Performance Management

- APM Input Tag will write the data by Writeback Datasource
- For APM Input Tag writeback to UniSim, Write tag name should have prefix **<PDT Input Name><.><Unisim Tag>**. Example **INPUT.BASE_AIR_MOLARFLOWRATE**
- UNISIM Output will read from APM Read Datasource

For Unisim Output tag read to APM, Overridden Tag name should have prefix **<PDT Output Name><. ><Unisim Tag>**. Example **OUTPUT.OP_ACT_AIRFANPOWER**

- Map the corresponding input and output tags in Asset Performance Management.

Below is the example for Tag mapping

Input from Unisim case

Object	Variable	Value	Units	Tag	Access Mode
Heater calcs	Cell Value (B-2)	19984.00195		BASE_AIR_MOLARFLOWRATE	Read/Write
Heater calcs	Cell Value (B-3)	1418.58667		BASE_FG_FLOWRATE	Read/Write
Heater calcs	Cell Value (B-12)	662.8		BASE_FG_CONVECTIONINLET_TEMP	Read/Write
Heater calcs	Cell Value (B-13)	438.6		BASE_FG_CONVECTIONOUTLET_TEMP	Read/Write
Heater calcs	Cell Value (B-24)	325.35		BASE_PROCESS_OUTLETTEMP	Read/Write
Heater calcs	Cell Value (B-29)	284.377045		BASE_PROCESS_INTLETTEMP	Read/Write
Heater calcs	Cell Value (B-33)	570		BASE_PRODUCT_FLOWRATE	Read/Write
Heater calcs	Cell Value (B-36)	1151		BASE_PRODUCT_OUTLETPRESSURE	Read/Write
Heater calcs	Cell Value (B-43)	250		BASE_AIR_EXITTEMP	Read/Write
Heater calcs-2	Cell Value (B-2)	19984.00195		BASE_AIR_MOLARFLOWRATE	Read/Write
Heater calcs-2	Cell Value (B-3)	1418.58667		BASE_FG_FLOWRATE	Read/Write
Heater calcs-2	Cell Value (B-12)	662.8		BASE_FG_CONVECTIONINLET_TEMP	Read/Write
Heater calcs-2	Cell Value (B-13)	438.6		BASE_FG_CONVECTIONOUTLET_TEMP	Read/Write
Heater calcs-2	Cell Value (B-24)	325.35		BASE_PROCESS_OUTLETTEMP	Read/Write
Heater calcs-2	Cell Value (B-29)	284.377045		BASE_PROCESS_INTLETTEMP	Read/Write
Heater calcs-2	Cell Value (B-33)	570		BASE_PRODUCT_FLOWRATE	Read/Write
Heater calcs-2	Cell Value (B-36)	1034.336792		BASE_PRODUCT_OUTLETPRESSURE	Read/Write
Heater calcs-2	Cell Value (B-43)	271.46		BASE_AIR_EXITTEMP	Read/Write

Input mapping in Asset Attribute Mapping Sheet

Name	Write DataSource	Write Tag Name
Base_Air_MolarFlowRate	USD	INPUT.BASE_AIR_MOLARFLOWRATE
Base_FG_FlowRate	USD	INPUT.BASE_FG_FLOWRATE
Base_FG_ConvectionInlet_Temp	USD	INPUT.BASE_FG_CONVECTIONINLET_TEMP
Base_FG_ConvectionOutlet_Temp	USD	INPUT.BASE_FG_CONVECTIONOUTLET_TEMP
Base_Process_OutletTemp	USD	INPUT.BASE_PROCESS_OUTLETTEMP
Base_Process_IntletTemp	USD	INPUT.BASE_PROCESS_INTLETTEMP
Base_Product_FlowRate	USD	INPUT.BASE_PRODUCT_FLOWRATE
Base_Product_OutletPressure	USD	INPUT.BASE_PRODUCT_OUTLETPRESSURE
Base_Air_ExitTemp	USD	INPUT.BASE_AIR_EXITTEMP
Base_Air_MolarFlowRate	USD	INPUT.BASE_AIR_MOLARFLOWRATE
Base_FG_FlowRate	USD	INPUT.BASE_FG_FLOWRATE
Base_FG_ConvectionInlet_Temp	USD	INPUT.BASE_FG_CONVECTIONINLET_TEMP
Base_FG_ConvectionOutlet_Temp	USD	INPUT.BASE_FG_CONVECTIONOUTLET_TEMP
Base_Process_OutletTemp	USD	INPUT.BASE_PROCESS_OUTLETTEMP
Base_Process_IntletTemp	USD	INPUT.BASE_PROCESS_INTLETTEMP

Base_Product_FlowRate	USD	INPUT.BASE_PRODUCT_FLOWRATE
Base_Product_OutletPressure	USD	INPUT.BASE_PRODUCT_OUTLETPRESSURE
Base_Air_ExitTemp	USD	INPUT.BASE_AIR_EXITTEMP

Similarly Output tags in Unisim case

Object	Variable	Value	Unit	Tag	Access Mode
K-100	Power	2.453	kW	OP_ACT_AIRFANPOWER	Read/Write
Heater calcs	Cell Value (F-19)	40.436	%	OP_ACT_EXCESSAIR	Read/Write
Heater calcs	Cell Value (F-20)	73.774		OP_ACT_OVERALLEFFICIENCY	Read/Write
FH-100	Total Duty	397183 30	kJ/h	OP_ACT_TOT_DUTY_PREDICT	Read/Write
Heater calcs	Cell Value (B-21)	14.087		ACT_AIRTOFUELRATIO	Read/Write
FH-100	Combustion Out Temperature	1833.9 07	C	OP_ACT_FLAMETEMP	Read/Write
Heater calcs	Cell Value (B-32)	12.867 45	kJ/h	ACT_FUELDUTY	Read/Write
E-106	Exchanger Cold Duty	297028 25	kJ/h	OP_ACT_RADIATIONDUTY	Read/Write
E-102	Exchanger Cold Duty	756707 6	kJ/h	OP_ACT_CONVECTIONDUTY	Read/Write
Heater calcs-2	Cell Value (F-19)	40.436	%	OP_EXP_EXCESSAIR	Read/Write
FH-100-2	Combustion Out Temperature	1762.2 7	C	OP_EXP_FLAMETEMP	Read/Write
Objective function-2	Cell Value (E-3)	725.97 4	C	OP_EXP_FLUEGASRADIATIONOUTTEMP	Read/Write
Objective function-2	Cell Value (E-5)	460.63 0	C	OP_EXP_CONVECTIONOUTTEMP	Read/Write
Objective function-2	Cell Value (E-6)	356.00 0	C	OP_EXP_PROCESSOUTTEMP	Read/Write
Heater calcs-2	Cell Value (G-37)	879539 8	kJ/h	OP_EXP_CONVECTIONDUTY	Read/Write
Heater calcs-2	Cell Value (G-38)	332993 25	kJ/h	OP_EXP_RADIATIONDUTY	Read/Write
Heater calcs-2	Cell Value (G-39)	420947 24	kJ/h	OP_ACT_TOT_DUTY_PREDICT	Read/Write
Heater calcs-2	Cell Value (G-40)	78.188 5		OP_EXP_EFFICIENCY_PREDICT	Read/Write
Heater calcs	Cell Value (K-41)	5.6405 0	mol %	ACT_STACKO2	Read/Write
Heater calcs	Cell Value (L-41)	5.6405	mol %	EXP_EXPECTEDSTACKO2	Read/Write

Heater calcs	Cell Value (K-42)	673675 2	kJ/h	ACT_APHDUTY	Read/Writ e
Heater calcs	Cell Value (L-42)	721627 7	kJ/h	EXP_EXPECTEDAPHDUTY	Read/Writ e
Heater calcs	Cell Value (K-43)	217.59 4	C	ACT_STACKTEMP	Read/Writ e
Heater calcs	Cell Value (L-43)	231.49 6	C	EXP_EXPECTEDSTACKTEMP	Read/Writ e
Heater calcs	Cell Value (K-44)	0.22		ACT_APHBYPASS	Read/Writ e
Heater calcs	Cell Value (L-44)	0.22		EXP_EXPECTEDAPHBYPASS	Read/Writ e
Heater calcs	Cell Value (L-46)	2.453	kW	OP_EXP_AIRFANPOWER	Read/Writ e

Output mapping in Asset Attribute Mapping Sheet

Name	Read DataSource	Overridden Tag Name
OP_ACT_AirFanPower	USD	OUTPUT.OP_ACT_AIRFANPOWER
OP_ACT_ExcessAir	USD	OUTPUT.OP_ACT_EXCESSAIR
OP_ACT_OverallEfficiency	USD	OUTPUT.OP_ACT_OVERALLEFFICIENCY
OP_ACT_TOT_DUTY_PREDICT	USD	OUTPUT.OP_ACT_TOT_DUTY_PREDICT
ACT_Airtofuelratio	USD	OUTPUT.ACT_AIRTOFUELRATIO
OP_ACT_FlameTemp	USD	OUTPUT.OP_ACT_FLAMETEMP
ACT_FuelDuty	USD	OUTPUT.ACT_FUELDUTY
OP_ACT_RadiationDuty	USD	OUTPUT.OP_ACT_RADIATIONDUTY
OP_ACT_ConvectionDuty	USD	OUTPUT.OP_ACT_CONVECTIONDUTY
OP_EXP_ExcessAir	USD	OUTPUT.OP_EXP_EXCESSAIR
OP_EXP_FlameTemp	USD	OUTPUT.OP_EXP_FLAMETEMP
OP_EXP_FlueGasRadiationOutTemp	USD	OUTPUT.OP_EXP_FLUEGASRADIATIONOUTTEMP
OP_EXP_CONVECTIONOutTemp	USD	OUTPUT.OP_EXP_CONVECTIONOUTTEMP
OP_EXP_ProcessOutletTemp	USD	OUTPUT.OP_EXP_PROCESSEOUTTEMP
OP_EXP_ConvectionDuty	USD	OUTPUT.OP_EXP_CONVECTIONDUTY
OP_EXP_RadiationDuty	USD	OUTPUT.OP_EXP_RADIATIONDUTY
OP_ACT_TOT_DUTY_PREDICT	USD	OUTPUT.OP_ACT_TOT_DUTY_PREDICT
OP_EXP_EFFICIENCY_PREDICT	USD	OUTPUT.OP_EXP_EFFICIENCY_PREDICT
ACT_StackO2	USD	OUTPUT.ACT_STACKO2
EXP_ExpectedStackO2	USD	OUTPUT.EXP_EXPECTEDSTACKO2
ACT_APHDuty	USD	OUTPUT.ACT_APHDUTY
EXP_ExpectedAPHDuty	USD	OUTPUT.EXP_EXPECTEDAPHDUTY
ACT_StackTemp	USD	OUTPUT.ACT_STACKTEMP
EXP_ExpectedStackTemp	USD	OUTPUT.EXP_EXPECTEDSTACKTEMP
ACT_APHBypass	USD	OUTPUT.ACT_APHBYPASS
EXP_ExpectedAPHBypass	USD	OUTPUT.EXP_EXPECTEDAPHBYPASS

OP_EXP_AirFanPower	USD	OUTPUT.OP_EXP_AIRFANPOWER
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2.1.8 APM-Unisim Calculation Scheduler

For Unisim execution to trigger, polling interval with desired frequency to be updated manually from DB server for External datasource.

1. Go to DB server
2. Expand the External Data sources table from Asset Sentinel database.
3. Edit the USD datasource by adding Polling interval as below

```
<ConnectionXml>
<Type>USD</Type>
<InputDirectory>\\10.11.34.215\USD File\Heater_MH_F_01_Rev02.usc</InputDirectory>
<Mode>0</Mode>
<OutputDirectory>\\10.11.34.215\USD File\Snapshots</OutputDirectory>
<Timeout>15</Timeout>
<PollingInterval>10</PollingInterval>
<Description>USD
EXPECT</Description>ConnectionStringProcessBatchIntervalMs=500;MaximumTasks=5;MaximumWaiting
Tasks=1;MaximumWaitingTimeSeconds=30;MinimumTagsPerBatch=0;MaximumTagsPerBatch=500;Minimu
mDataPointsPerTag=30;BatchStatisticsSeconds=60;UpdateCountersIntervalMs=1000;WriteFailureRetries=5
</ConnectionXml>
```

4. Once done Take AM service restart to effect polling interval and USD datasource for “Running”.

2.1.9 Data in Attribute Overview

Once AM service get resumed, output data from Unisim get written for the attributes with expected frequency as shown below in Attribute Overview page.

System Console MH-F-01 Expected Heater USD Model ⓘ Attribute Overview MH-F-01 Expected Heater USD Model ⓘ							
🔍 🔍 🔍 🔍 🔍 🔍 🔍 🔍 Search Asset or Attributes 56 total row(s)							
Quality	Attribute	Value	Unit	Low Limit	Historical	Timestamp	
Good	Actual Air to Fuel Ratio	10.119				11/11/2024 04:18:03 PM	
Good	Actual APH Duty	3.1085e+6				11/11/2024 04:18:03 PM	
Good	Actual Convection Duty	9.3321e+6				11/11/2024 04:18:03 PM	
Good	Actual Convection Out Temp	392.96				11/11/2024 04:18:03 PM	
Good	Actual Excess Air	0.87411				11/11/2024 04:18:03 PM	
Good	Actual Fan Power	1.7190				11/11/2024 04:18:03 PM	
Good	Actual Flame Temp	2112.1				11/11/2024 04:18:03 PM	
Good	Actual Fuel Duty	39.274				11/11/2024 04:18:03 PM	
Good	Actual Overall Efficiency	81.745				11/11/2024 04:18:03 PM	
Good	Actual Radiation Duty	2.3337e+7				11/11/2024 04:18:03 PM	
Good	Actual Stack O2	0.16524				11/11/2024 04:18:03 PM	
Good	Actual Stack Temperature	275.75				11/11/2024 04:18:03 PM	
Good	Actual Total Duty	4.2935e+7				11/11/2024 04:18:03 PM	

2.1.10 Problem:

2.1.10.1 Incorrect mapping in APM-Unisim Input and output variable.

- If no data or incorrect data get written for attributes in the APM, check the AM log in calculation server with search as USD.
- Check the error for any attribute “not found in USD datasource” in the AM log.
- Correct the Attributes for its mapping in Unisim case or remove the unwanted attributes which is not required for APM-Unisim.

2.1.10.2 In UAS 540.1, APM to Unisim (R510) Integration is not working.

Below pre-requisite to be done while creating the interface.

1. DCOM setting update:

- Add AMUsersgroup in Distributed COM Users Group
- Go to *Start > All Apps > Component Services > Computers > My Computer* and right click properties > COM Security and Add **NETWORK** in **Access Permissions** for Edit Limits and Edit Default **Launch and Activation Permissions** for Edit Limits and Edit Default. Give all permissions for **NETWORK**.

- Go to *Start > All Apps > Component Services > Computers > My Computer>DCOM Config> OPCEnum* and right click properties > Security and mark “Customize” for Launch and Activation Permission, Access Permission and Configuration Permission.
- USD datasource re-create – Since existing UNISIM datasource was not get displayed in the calc status for calculation schedule, created additional USD datasource and removed earlier UNISIM datasource.
 - Unisim Process License is required to get Snapshots for Input file into Fail and success folder.
 - Runtime account (Service Account) and AMusersgroup permission for Unisim (usc) file – Right click on UniSim folder (at Calc Server) properties, add Everyone, Service account and AMusersgroup with full control and **Enable inheritance** for all the users from Advance.
 - Polling interval with desired frequency to be updated manually from DB server for External datasource

```
<ConnectionXml>
<Type>USD</Type>
<InputDirectory>\\10.11.34.215\USD File\Heater_MH_F_01_Rev02.usc</InputDirectory>
<Mode>0</Mode>
<OutputDirectory>\\10.11.34.215\USD File\Snapshots</OutputDirectory>
<Timeout>15</Timeout>
<PollingInterval>10</PollingInterval>
<Description>USD
EXPECT</Description>ConnectionStringProcessBatchIntervalMs=500;MaximumTasks=5;MaximumWaiting
Tasks=1;MaximumWaitingTimeSeconds=30;MinimumTagsPerBatch=0;MaximumTagsPerBatch=500;Minimu
mDataPointsPerTag=30;BatchStatisticsSeconds=60;UpdateCountersIntervalMs=1000;WriteFailureRetries=5
</ConnectionXml>
```

- Mappings of attribute between Unisim case and UAS asset attribute mappings are streamlined.
- Unisim PDT of Input and Output should not have blank row with No Transfer.

INPUT Data

	Object	Variable	Value	Units	Tag	Access Mode
1	Heater calcs	Cell Value (B-2)	1.947e+004		BASE_AIR_MOLARFLOWRATE	Read/Write
2	Heater calcs	Cell Value (B-3)	1440		BASE_FG_FLOWRATE	Read/Write
3	Heater calcs	Cell Value (B-12)	670.6		BASE_FG_CONVECTIONINLET	Read/Write
4	Heater calcs	Cell Value (B-13)	436.4		BASE_FG_CONVECTIONOUTLET	Read/Write
5	Heater calcs	Cell Value (B-24)	325.6		BASE_PROCESS_OUTLETTEMP	Read/Write
6	Heater calcs	Cell Value (B-29)	284.8		BASE_PROCESS_INLETTEMP	Read/Write
7	Heater calcs	Cell Value (B-33)	549.1		BASE_PRODUCT_FLOWRATE	Read/Write
8	Heater calcs	Cell Value (B-36)	1015		BASE_PRODUCT_OUTLETTEMP	Read/Write
9	Heater calcs	Cell Value (B-43)	274.1		BASE_AIR_EXITTEMP	Read/Write

OUTPUT Data

	Object	Variable	Value	Units	Tag	Access Mode
1	K-100	Power	2.389	kW	OP_ACT_AIRFANPOWER	Read/Write
2	Heater calcs	Cell Value (F-19)	34.79	%	OP_ACT_EXCESSAIR	Read/Write
3	Heater calcs	Cell Value (F-20)	69.76		OP_ACT_OVERALLEFFICIENCY	Read/Write
4	FH-100	Total Duty	3.812e+007	kJ/h	OP_ACT_TOT_DUTY_PREDICT	Read/Write
5	Heater calcs	Cell Value (B-21)	13.52		ACT_AIRTOFUELRATIO	Read/Write
6	FH-100	Combustion Out Temp	1815	C	OP_ACT_FLAMETEMP	Read/Write
7	Heater calcs	Cell Value (B-32)	13.06	kJ/h	ACT_FUELDUTY	Read/Write
8	E-106	Exchanger Cold Duty	3.259e+007	kJ/h	OP_ACT_RADIATIONDUTY	Read/Write
9	E-102	Exchanger Cold Duty	8.155e+006	kJ/h	OP_ACT_CONVECTIONDUTY	Read/Write
10	Objective func	Cell Value (E-3)	719.9	C	OP_EXP_FLUEGASRADIATIONK	Read/Write
11	Objective func	Cell Value (E-5)	437.0	C	OP_EXP_CONVECTIONOUTLET	Read/Write
12	Objective func	Cell Value (E-6)	356.9	C	OP_EXP_PROCESSOUTTEMP	Read/Write
13	Heater calcs-2	Cell Value (G-37)	9.158e+006	kJ/h	OP_EXP_CONVECTIONDUTY	Read/Write
14	Heater calcs-2	Cell Value (G-38)	3.453e+007	kJ/h	OP_EXP_RADIATIONDUTY	Read/Write
15	Heater calcs-2	Cell Value (G-39)	4.369e+007	kJ/h	OP_EXP_TOT_DUTY_PREDICT	Read/Write
16	Heater calcs-2	Cell Value (G-40)	79.96	mol %	OP_EXP_OVERALLEFFICIENCY	Read/Write
17	Heater calcs	Cell Value (K-41)	5.042	mol %	ACT_STACKO2	Read/Write
18	Heater calcs	Cell Value (L-41)	5.042	mol %	EXP_EXPECTEDSTACKO2	Read/Write
19	Heater calcs	Cell Value (K-42)	6.685e+006	kJ/h	ACT_APHDUTY	Read/Write
20	Heater calcs	Cell Value (L-42)	6.081e+006	kJ/h	EXP_EXPECTEDAPHDUTY	Read/Write
21	Heater calcs	Cell Value (K-43)	190.2	C	ACT_STACKTEMP	Read/Write
22	Heater calcs	Cell Value (L-43)	239.4	C	EXP_EXPECTEDSTACKTEMP	Read/Write
23	Heater calcs	Cell Value (K-44)	-0.4586		ACT_APHBYPASS	Read/Write
24	Heater calcs	Cell Value (L-44)	0.2200		EXP_EXPECTEDAPHBYPASS	Read/Write
25	Heater calcs	Cell Value (L-46)	2.389	kW	OP_EXP_AIRFANPOWER	Read/Write

- Unisim case is corrected with Warning by normalizing the composition.
- All input values from APM are updated with latest timestamp and followed by AM Service restart.